Question (3): Best Rank-1 Approximation of A^2

We are working with a matrix A, which is expressed in the form $A = PDP^T$. Let's first understand what this means and then solve the problem step by step.

Step 1: What is a Matrix?

A matrix is like a table of numbers arranged in rows and columns. For example:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}.$$

Here:

- The rows are horizontal (e.g., $[1, 2]$).
- The columns are vertical (e.g., $[1, 3]$).

Matrices are used to represent data or operations in math.

Step 2: What Does $A = PDP^T$ Mean?

This equation tells us something special about A:

1. **Matrix P:**
   - P is a matrix that holds information about the **eigenvectors** (directions) of A.

- In this problem, a specific version of P^{-1} is given as:

$$P^{-1} = \begin{bmatrix} 1 & 0 & \dots & 0 \\ 0 & \frac{1}{2} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \frac{1}{m} \end{bmatrix}.$$

2. **Matrix D:**
   - D is a **diagonal matrix**, meaning all the numbers are on the diagonal, and the rest are zeros:

$$D = \mathrm{diag}(1, l^2, l^3, \dots, l^m),$$

   where l is a constant.
3. **Transpose P^T:**
   - P^T is the transpose of P, which just means flipping the rows and columns of P.

In simple terms, $A = PDP^T$ means that A can be broken down into three simpler parts: a diagonal matrix D, and matrices P and P^T that help rotate or transform the data.

Step 3: What is A^2?

To find A^2, we multiply A by itself:

$$A^2 = A \cdot A.$$

Using the form $A = PDP^T$, we compute:

$$A^2 = (PDP^T)(PDP^T).$$

Since $P^T P = I$ (the identity matrix, which doesn't change anything when multiplied), this simplifies to:

$$A^2 = PD^2P^T.$$

Step 4: What is D^2?

To calculate D^2, we square all the diagonal entries of D:

$$D = \text{diag}(1, l^2, l^3, \dots, l^m).$$

Squaring each entry gives:

$$D^2 = \text{diag}(1, l^4, l^6, \dots, l^{2m}).$$

Step 5: What is a Rank-1 Approximation?

A **rank-1 matrix** is a very simple matrix that can be written as:

$$uv^T,$$

where u and v are column vectors. It's like summarizing a big matrix using only the most important direction.

To find the **best rank-1 approximation** of A^2:

1. We look for the **largest eigenvalue** in D^2, which is:

$$l^{2m}.$$

2. The vector associated with this eigenvalue (the eigenvector) is the **last column of P**, called p_m.

3. The rank-1 approximation is:

$$A^2 \approx l^{2m} p_m p_m^T.$$

Final Answer for Question (3):

The best rank-1 approximation of A^2 is:

$$l^{2m} p_m p_m^T.$$

Question (ii): Find a 2×2 Matrix A

We need to find a 2×2 matrix A such that:

1. When A multiplies $v_1 = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}$, the result is:

$$Av_1 = -v_1.$$

2. When A multiplies $v_2 = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{bmatrix}$, the result is:

$$Av_2 = -v_2.$$

Step 1: Represent the Matrix A

Let:

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}.$$

When A multiplies a vector $v = \begin{bmatrix} x \\ y \end{bmatrix}$, the result is:

$$Av = \begin{bmatrix} ax + by \\ cx + dy \end{bmatrix}.$$

Step 2: Apply the Condition for v_1

Substitute $v_1 = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}$ into $Av_1 = -v_1$:

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix} = - \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}.$$

This gives two equations:

1. $a \cdot \frac{1}{\sqrt{2}} + b \cdot \frac{1}{\sqrt{2}} = -\frac{1}{\sqrt{2}}$, which simplifies to:

$$a + b = -1. \tag{1}$$

2. $c \cdot \frac{1}{\sqrt{2}} + d \cdot \frac{1}{\sqrt{2}} = -\frac{1}{\sqrt{2}}$, which simplifies to:

$$c + d = -1. \tag{2}$$

Step 3: Apply the Condition for v_2

Substitute v_2 = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} into $Av_2 = -v_2$:

$$[\begin{matrix} a & b \\ c & d \end{matrix}][\begin{matrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{matrix}] = -[\begin{matrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{matrix}].$$

This gives two equations:

1. $a \cdot \frac{1}{\sqrt{2}} - b \cdot \frac{1}{\sqrt{2}} = -\frac{1}{\sqrt{2}}$, which simplifies to:

$$a - b = -1. \tag{3}$$

2. $c \cdot \frac{1}{\sqrt{2}} - d \cdot \frac{1}{\sqrt{2}} = \frac{1}{\sqrt{2}}$, which simplifies to:

$$c - d = 1. \tag{4}$$

Step 4: Solve the System of Equations

From (1) and (3):

- $a + b = -1$,
- $a - b = -1$.

Add the equations:

$$2a = -2 \implies a = -1.$$

Subtract the equations:

$$2b = 0 \implies b = 0.$$

From (2) and (4):

- $c + d = -1$,
- $c - d = 1$.

Add the equations:

$$2c = 0 \implies c = 0.$$

Subtract the equations:

$$2d = -2 \implies d = -1.$$

Step 5: Write the Matrix A

The matrix A is:

$$A = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}.$$

Final Answers

1. **Question (3)**: The best rank-1 approximation of A^2 is:

$$l^{2m} p_m p_m^T.$$

2. **Question (ii)**: The 2×2 matrix A is:

$$A = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}.$$